ASBESTOS SURVEY

At

ALBUQUERQUE SERVICE UNIT
Building No. 21

For

Albuquerque Area Indian Health Service
Division of Health Facilities
5300 Homestead Road, NE
Albuquerque, New Mexico 87110

By

U.S. Public Health Service
Federal Occupational Health
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June 2007

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Area Wide Asbestos Building Inspection
Indian Health Service Facilities
Albuquerque Service Unit
Building No. 21
Albuquerque, New Mexico

1.0 INTRODUCTION

As requested by the Albuquerque Area Indian Health Service, bulk sampling of suspect asbestos-containing building materials was conducted in the original 1930 section and the 1980 addition of Building No. 21, located at the Albuquerque Service Unit, Albuquerque, New Mexico, on April 14, 2006. Bulk samples of suspect asbestos-containing materials (ACMs) were collected throughout the interior and exterior of the building.

Ms. Stephanie Jury, representing Federal Occupational Health (FOH), performed the asbestos inspection of the subject property. Ms. Jury holds current Asbestos Hazard Emergency Response Act (AHERA) accreditation as an Asbestos Building Inspector. Sampling of suspect asbestos-containing materials was conducted using procedures similar to the Environmental Protection Agency (EPA) AHERA regulations, 40 CFR 763. The Federal Occupational Health EH Environmental Laboratory in Denver, Colorado, performed the laboratory analysis of the asbestos samples.

Laboratory analysis indicates that various floor tiles and mastics, sheet vinyl flooring mastic, sink undercoating, interior and exterior window/door frame caulk, fire doors, fire door frames, exterior stucco, and roof penetration mastic in the original 1930 section of Building No. 21 contain asbestos. Laboratory analysis also indicates that floor tile mastic, sheet vinyl flooring, sink undercoating, and cements asbestos panels in the 1980 addition contain asbestos. Fire doors and fire door frames throughout the 1980 addition are assumed to be positive for asbestos.

2.0 DESCRIPTION OF INSPECTED AREA

This inspection focused on the original 1930 section and the 1980 addition of the entire interior and exterior of the building. The 1930 section of the building is four stories with a penthouse and a stucco exterior built slab-on grade. The roof consists of a membrane roofing system with various penetration mastics. The 1980 addition is two stories built slab on grade with a stucco exterior and a membrane roof.

Suspect building materials identified and sampled in the subject property included the following: various floor tiles and associated mastics, sheet vinyl flooring and associated mastics, carpet mastic, various floor and wall ceramic tile grout, various gypsum drywall materials, plaster materials, wall sheet vinyl, concrete masonry unit (CMU), CMU mortar, various bricks, brick mortar, various ceiling tiles, fireproofing, fireproof caulk, fire doors, fire door frames, exterior stucco, various covebase mastics, sink undercoating, interior/exterior window and door frame caulk, exterior window putty, various caulks, cement asbestos panels and various roof mastics.

3.0 SURVEY PROCEDURES

Each room was assigned unique functional space numbers. Each functional space was visually inspected for suspect asbestos-containing materials. Materials were mapped and quantified on maps provided by the IHS or hand drawn maps.

Each suspect ACM was designated as a distinct homogeneous area, which is a single material, uniform in texture and appearance, installed at one time, and unlikely to consist of more than one type or formulation of material. The inspector touched each suspect material in order to determine friability.

A sufficient number of samples were collected of each material to satisfy the Occupational Safety and Health Administration (OSHA) and the National Emission Standard for Hazardous Air Pollutants

(NESHAP) regulations for the determination of asbestos content. A total of 152 samples were collected from the original 1930 section. 71 samples were collected from the 1980 addition of Building 21.

Each sample was assigned a unique sample identification number and assessed for damage and friability classifications. The samples, with Chain of Custody documentation were delivered to FOH EH Environmental Laboratory for analysis. The FOH EH Environmental Laboratory is accredited under the National Voluntary Laboratory Accreditation Program (NVLAP) for bulk-asbestos sample analysis conducted by the National Institute of Standards and Technology.

A description of the samples taken, location, and summary of laboratory results are located in Appendix 1. The asbestos laboratory reports are included as Appendix 2.

4.0 FINDINGS

Results of the asbestos inspection are as follows:

Table 1:
Albuquerque Service Unit
Building No. 21
Original 1930 Section
Bulk Sample Results
April 14, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	12”x12” Floor Tile & Mastic (White w/ Gray)	102, 104-111, 127-132, 122-125, 119D, C101, C102, C104, C105, S102, 201, 202, 204-206, 209, 210, 224, 226, 228, 229, 230, 232, 233, 234, C201-C203, E-204, S201, C301,301-311, 321-323, C302, C304, S302	No	Good	16,756 sf
1B	**12”x12” Floor Tile & Mastic (Lt. Green)**	**114, 121, C105A, 313, 313B, 314, 314B, 315, 316, 318, 318A, 319, 320, C303, 401, 404, 406, 410-416, 418-421, C403, C404**	**Yes: 5-10% Chrysotile**	**Good**	**5,300 sf**
1C	12”x12” Floor Tile & Mastic (Lt. Brown)	116, C-104	No	Good	700 sf
1D	Sheet Vinyl Flooring (Blue/White)	115, 117	No	Good	268 sf
1E	**12”x12” Floor Tile & Mastic (Green/Dark)**	**117, 118**	**Yes: 10-15% Chrysotile**	**Good**	**450 sf**
1F	Ceramic Tile Grout	230B	No	Good	150 sf
1G	**Sheet Vinyl Flooring Gray**	**201, 208, 211**	**Yes: 5-10% Chrysotile (mastic only)**	**Good**	**450 sf**

HA #	Material	Functional Space	ACM	Condition	Quantity
1H	12”x12” Floor Tile & Mastic (Green/Brown/Gray)	121, C105A	No	Good	152 sf
1I	12”x12” Floor Tile & Mastic (Blue/Black)	203, 204, 214-217, 217A, 218-220, 220A, 221, 222, 235, 313B, 314,314-316A, 318B, 405, 407, 408, 413A, 415A, 416A, 418A, 419A	No	Good	4,252 sf
1J	**12”x12” Floor Tile & Mastic (Beige)**	**C404A**	**Yes: 3-6% Chrysotile (mastic only)**	**Good**	**100 sf**
1K	White Ceramic Tile Grout (Gray)	402A, 407A, 408A, 410A	No	Good	400 sf
2A	Gypsum Drywall	Throughout	No	Good	35,400 sf
2B	Joint Compound	Throughout	No	Good	35,400 sf
2C	Drywall Texture 1 (Bumpy)	104, 105, 106A, 108, 124, 125, 125A, 125C, 129, 129A&B, 130, 131, C105A, 224A, 321A, 401, 402, 405-408, 408A, 410, 410A, 412-422, C403, C404, C404A	No	Good	10,784 sf
2D	Drywall Texture 2 (Thick Orange Peel)	130	No	Good	200 sf
2E	Drywall Texture 3 (Thick Blotchy)	132, 132A	No	Good	224 sf
2F	Plaster Skin Coat	102, 104B, 106, 109-113, 113A, 119, 119A&B, 121-127, 130, C101, C102, C105, S101, S102, 201-205, 205B, 206-211, 226, 226A, 227-229, 230A, 232, 233, C201, C202, C203, 301, 301A, 303-309, 309B, 310, 311, 316, 320, 321, 323, C301, C302, C304, S302, 402, 404, 411, 415, 416, 420, C403, 501	No	Good	70,363 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
2G	Plaster Brown Coat	102, 104B, 106, 109-113, 113A, 119, 119A&B, 121-127, 130, C101, C102, C105, S101, S102, 201-205, 205B, 206-211, 226, 226A, 227-229, 230A, 232, 233, C201, C202, C203, 301, 301A, 303-309, 309B, 310, 311, 316, 320, 321, 323, C301, C302, C304, S302, 402, 404, 411, 415, 416, 420, C403, 501	No	Good	70,363 sf
2H	4”x4” Ceramic Wall Tile Grout (White)	230, 402A, 407A, 408A,410A 415A, 416A, 421, 422	No	Good	1,014 sf
2I	Wall Tile Grout (Gray)	C201	No	Good	40 sf
2J	CMU	108A, 502	No	Good	1,000 sf
2K	CMU Mortar	108A, 502	No	Good	1,000 sf
2L	Orange Brick Mortar	112, 115, 116, 117, 118, 119,119A, 119D, 120, 120A	No	Good	8,000 sf
2M	Orange Brick	112, 115, 116, 117, 118, 119,119A, 119D, 120, 120A	No	Good	8,000 sf
2N	**Exterior Stucco**	**Exterior**	**Yes: 2-4% Chrysotile**	**Good**	**150,000 sf**
2O	Wall Sheet Vinyl	C102, C105, C202, C203	No	Good	2,550 sf
2P	Exterior Green Block	Exterior	No	Good	500 sf
3A	2’x4’ Fissured Ceiling Tile	102, 104-111, 116-119, 119D, 120, 122-125, 127-132, C101, C102, C104, C105, 201-206, 209-211, 214-222, 224A, 226, 226A, 228-230, 230A&B, 232-235, C201-C203, 301-310, 313-318, 320-323, 323B, C301-C304, C304A, 401, 402, 402A, 404, 406-408, 410, 410A, 411, 413, 413A, 415-419,421, 422, C403, C404, C404A	No	Good	50,000 sf
3B	2’x2’ Pinhole Ceiling Tile	C201	No	Good	400 sf
3C	2’x4’ Drywall Ceiling Tile	203	No	Good	24 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
6A	Fireproofing (Gray)	119D, 120, 120A,	No	Good	10,000 sf
6B	Red Fireproofing Caulk	501	No	Good	500 sf
6C	**Fire Doors**	**C101, S101, C102, S102, C104, C105, 109, 112, 119, 119D, C210, S201, S202, C203, 211, 224A, 230, 301, C301, S302, 303, 308, 309, 317, 323B, S403, C404, 405**	**Yes: 1-2% Chrysotile 5-10% Amosite**	**Good**	**37 Doors**
6D	Fire Door Frame	**C101, S101, C102, S102, C104, C105, 109, 112, 119, 119D, C210, S201, S202, C203, 211, 224A, 230, 301, C301, S302, 303, 308, 309, 317, 323B, S403, C404, 405**	No	Good	37 Door Frames
7A	Covebase Mastic (yellow)	Throughout Except: C101.C102, C105 501, 502, 503, 504	No	Good	1,316 sf
7B	Window/Door Caulking (White)	Throughout	No	Good	300 sf
7C	**Sink Undercoating**	**129, 218, 229, 232, 233, 404**	**Yes: 4-8% Chrysotile**	**Good**	**10 sinks**
7D	Exterior Window Putty	Exterior	No	Good	20 sf
7E	Covebase Mastic (Brown)	C101, C102, C105	No	Good	200 sf
7F	**Interior Door/ Window Caulk**	**Throughout Penthouse**	**Yes: 1-2% Chrysotile**	**Good**	**10 sf**
7G	Vent Caulk	502	No	Good	5 sf
7H	**Exterior Door/ Window Caulk**	**Exterior Penthouse**	**Yes: 2-4% Chrysotile**	**Good**	**5 sf**
7I	Exterior Green Block Caulk (White)	Roof	No	Good	200 sf
8A	Roof Penetration Mastic	Roof	No	Good	400 sf
8B	**Silver Roof Penetration Mastic**	**Roof**	**Yes: 10-20% Chrysotile**	**Damaged**	**200 sf**

Albuquerque Service Unit
Building No. 21
1980 Addition
April 14, 2006

HA #	Material	Functional Space	ACM	Condition	Quantity
1A	5”x5” Orange Ceramic Tile Grout	53	No	Good	950 sf
1B	12”x12” Floor Tile & Mastic (White w/ Blue)	50	No	Good	270 sf

HA #	Material	Functional Space	ACM	Condition	Quantity
1C	**12”x12” Floor Tile & Mastic (White w/ Brown)**	**Throughout except: 18A, 19, 20, 22, 25, 30, 31, 36, 48, 49, 50, 51, 52, 53 45A,**	**Yes: 3-6% Chrysotile (mastic only)**	**Good**	**17,305 sf**
1D	Ceramic Tile Grout	22, 31, 36, 45, 48, 49	No	Good	1,448 sf
1E	**Sheet Vinyl Flooring (Beige)**	**19, 20, 23**	**Yes: 10-20% Chrysotile**	**Good**	**900 sf**
1F	Carpet Mastic	27	No	Good	3,400 sf
2A	Gypsum Drywall	Throughout	No	Good	30,000 sf
2B	Joint Compound	Throughout	No	Good	30,000 sf
2C	Drywall Texture 1 (Bumpy)	Throughout Walls	No	Good	30,000 sf
2D	Wall Tile Grout	18A, 22, 30, 31, 36, 45A, 48, 49	No	Good	3,104 sf
2E	CMU	50, 52	No	Good	800 sf
2F	CMU Mortar	50, 52	No	Good	800 sf
2G	Exterior Stucco	Exterior	No	Good	38,200 sf
3A	2’x4’ Pinhole Ceiling Tile	18, 20, 25, 27, 29, 37, 45, 46	No	Good	18,000 sf
3B	2’x4’ Fissured Ceiling Tile	48, 49	No	Good	280 sf
6A	**Fire Doors**	**Throughout**	**Assumed**	**Good**	**60 doors**
6B	**Fire Door Frames**	**Throughout**	**Assumed**	**Good**	**60 door frames**
7A	Covebase Mastic (Yellow/Brown)	50	No	Good	700 sf
7B	**Sink Undercoating**	**26, 30, 31**	**Yes: 2-4% Chrysotile**	**Good**	**5 sinks**
7C	Interior Window Caulk (Gray)	Throughout	No	Good	100 sf
7D	**Cement Asbestos Panels**	**S103, S203, S303, S403**	**Yes: 25-35% Chrysotile**	**Good**	**150 sf**
8A	Black Roof Penetration Mastic	Roof	No	Good	200 sf
8B	White Roof Penetration Mastic	Roof	No	Good	200 sf

Laboratory analysis indicates that various floor tiles and mastics, sheet vinyl flooring mastic, sink undercoating, interior and exterior window/door frame caulk, fire doors, fire door frames, exterior stucco, and roof penetration mastic in the original 1930 section Building No. 21 contain asbestos. Laboratory analysis also indicates that floor tile mastic, sheet vinyl, sink undercoating, cement asbestos panels, fire doors, and fire door frames in the 1980 addition contain asbestos.

Floor tile is classified as a non-friable NESHAP Category I material. This material is currently in good condition and intact floor tile poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed flooring material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Floor tile mastic is classified as a non-friable NESHAP Category I material. This material is currently in good condition. Floor tile mastic that is beneath floor tile, i.e., inaccessible, poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed flooring material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Sheet vinyl flooring and associated materials are classified as friable NESHAP Regulated Asbestos-Containing Material (RACM). The material is currently in good condition and intact sheet vinyl flooring poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed sheet vinyl flooring and associated materials should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Sink undercoating is classified as non-friable NESHAP Category II material. The material is currently in good condition and undamaged sink undercoating poses little to no health threat. Removal of this material typically involves the physical removal of the sink and should be performed by appropriately trained and certified personnel. Removed materials should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Cement-asbestos materials are classified as non-friable NESHAP Category II material. The material is currently in good condition and undamaged cement-asbestos poses little to no health threat. Removal of this material should be performed by appropriately trained and certified personnel using intact removal methods. Removed materials should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Exterior stucco is classified friable NESHAP Regulated Asbestos-Containing Material (RACM). Stucco is an exterior material; therefore, building occupants will likely not be affected. The material is currently in good condition and undamaged stucco poses little to no health threat. Removal of this material should be performed by appropriately trained and certified personnel. Removed materials should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

In an intact condition, asbestos-containing fire doors are classified as non-friable NESHAP Category II material. Asbestos material is inside the doors, i.e., inaccessible; therefore, intact fire doors pose little to no health threat. Removal or disturbance (e.g. installing hardware) to the doors should be performed by appropriately trained and certified personnel. If removal is required, the doors should be treated as friable NESHAP Regulated Asbestos-Containing Material (RACM). Removed fire doors should be disposed at a landfill that is approved by the Environmental Protection Agency (EPA) for acceptance of friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Window and door frame caulking is classified as non-friable NESHAP Category II material. The material is currently in good condition and intact window frame caulking poses little to no health threat. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed caulking material should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

Roof penetration mastic is classified as non-friable NESHAP Category I material. The penetration mastic is an exterior material; therefore, building occupants will likely not be affected. Removal or disturbance to this material should be performed by appropriately trained and certified personnel. Removed penetration mastic should be disposed at a landfill that accepts non-friable asbestos waste. Landfills should be checked for acceptance of this material prior to disposal.

5.0 ABATEMENT COST ESTIMATE

The programming cost estimate for the removal of asbestos-containing materials follows below.

Identified asbestos-containing materials (ACMs) for the original 1930 section of Building 21 consist of: floor tile (5,750 sf), floor tile mastic (5,850 sf), sheet vinyl flooring mastic (450 sf), exterior stucco (150,000 sf), fire doors and frames (37doors/frames), sink undercoating (10 sinks), interior/exterior window/door caulk (15 sf) and roof penetration mastic (200 sf). Identified asbestos-containing materials (ACMs) for the 1980 addition of Building 21 include: floor tile mastic (17,305 sf), sheet vinyl flooring (900 sf), sink undercoating (5 sinks) cement asbestos panel (150 sf) fire doors (60 doors), and fire door frames (60).

This programming cost estimate was derived by use of the Means Cost Guide and by consultation with experienced local contractors. It reflects current industry pricing. Future fluctuations in industry pricing, both natural and seasonal, may affect this estimate. In addition, Service Units outside of metropolitan areas may incur additional costs due to travel requirements on the part of abatement contractors. For purposes of this estimate, an 18% mark-up for logistical costs is included.

This estimate is for abatement contractor costs only. Additional costs, which are not included in this estimate, are those of an environmental consultant and replacement of materials that have been removed.

Table 2:
Albuquerque Service Unit
Building No. 21
Original 1930 Section
Programming Cost Estimate

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile & Mastic	5,850 sf	$2.50	18%	$17,257.50
Sheet Vinyl Flooring Mastic	450 sf	$2.50	18%	$1,327.50
Sink Undercoating	10 sinks	$200.00	18%	$2,360.00
Exterior Stucco	150,000 sf	$3.00	18%	$531,000.00
Interior Exterior Window/Door Caulk	15 sf	$8.00	18%	$ 141.60
Roof Penetration Mastic	200 sf	$21.00	18%	$4,956.00
Fire Doors and Fire Door Frames	37	$60.00	18%	$2,619.60
Total				$559,662.20

Table 3:
Albuquerque Service Unit
Building No. 21
1980 Addition
Programming Cost Estimate

Asbestos-Containing Material	Quantity of Material (sf)	Unit Cost (per sf)	Logistical Markup	Price
Floor Tile Mastic	17,305 sf	$2.50	18%	$51,049.75
Sheet Vinyl Flooring	900 sf	$2.50	18%	$2,655.00
Sink Undercoating	5 sinks	$200.00	18%	$1,180.00
Cement Asbestos Panels	150 sf	$4.00	18%	$ 708.00
Fire Doors and Fire Door Frames	60	$60.00	18%	$4,248.00
Total				$59,840.75

The grand total cost estimate for Building 21 is $617,349.45.

6.0 PROJECT LIMITATIONS

This Project was performed using, as a minimum, practices consistent with standards acceptable within the industry at this time, and a level of diligence typically exercised by industrial hygiene and environmental consultants performing similar services.

The procedures used in this investigation attempt to establish a balance between the competing goals of limiting investigative and reporting costs and time, and reducing the uncertainty about unknown conditions. Therefore, because the findings of this report were derived from the scope, costs, time, and other limitations, the conclusions should not be construed as a guarantee that all environmental or occupational hazards have been identified and fully evaluated. Where sample collection and testing have been performed, FOH's professional opinions are based in part on the interpretation of data from discrete sampling locations that may not represent conditions at non-sampled locations. FOH assumes no responsibility for omissions or errors resulting from inaccurate information, or data, provided by sources outside of FOH or from omissions or errors in public records.

Furthermore, it is emphasized that the final decision on how much risk to accept always remains with the client since FOH is not in a position to fully understand all of the client's needs. Clients with a greater aversion to risk may want to take additional actions while others, with less aversion to risk, may want to take no further action

APPENDIX 1
Location of Functional Spaces
Bulk Sample Log

APPENDIX 2
Laboratory Results

APPENDIX 3
Certifications